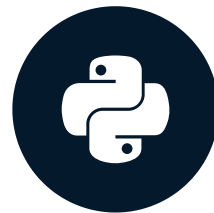


Welcome!

BUILDING AGENTIC WORKFLOWS WITH LLAMAINDEX

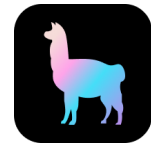


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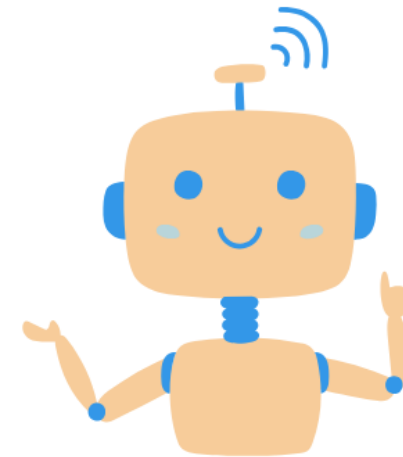
What are we building?

- Smart agentic workflows for research in



LlamaIndex

- Planning



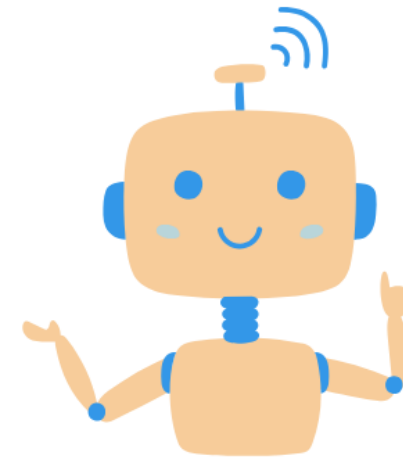
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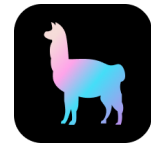
LlamaIndex

- Planning
- Use tools



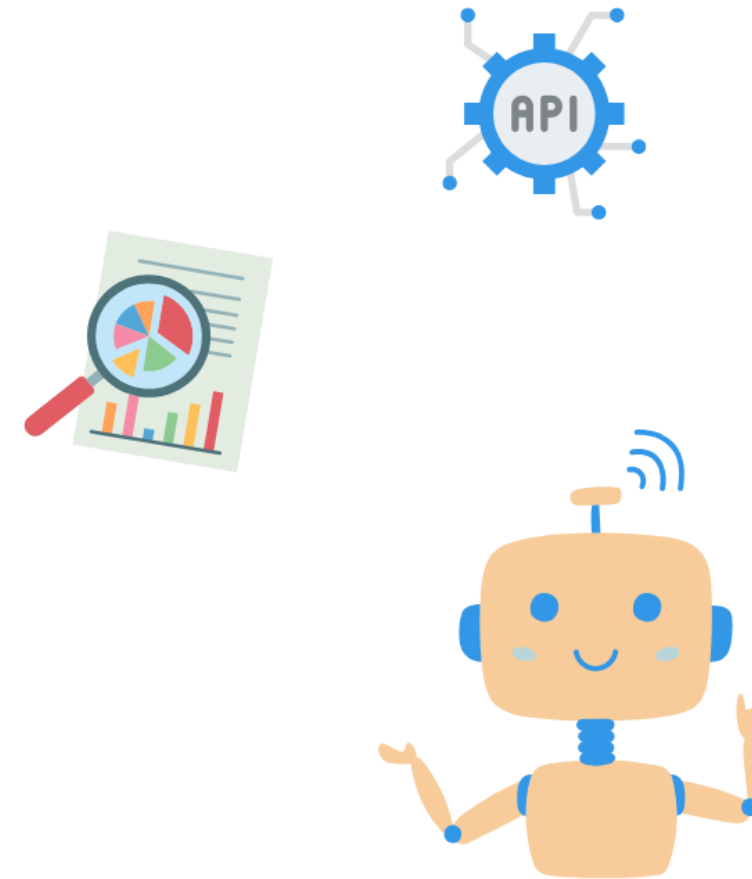
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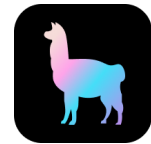
LlamaIndex

- Planning
- Use tools
- Remember information



What are we building?

- Smart agentic workflows for research in



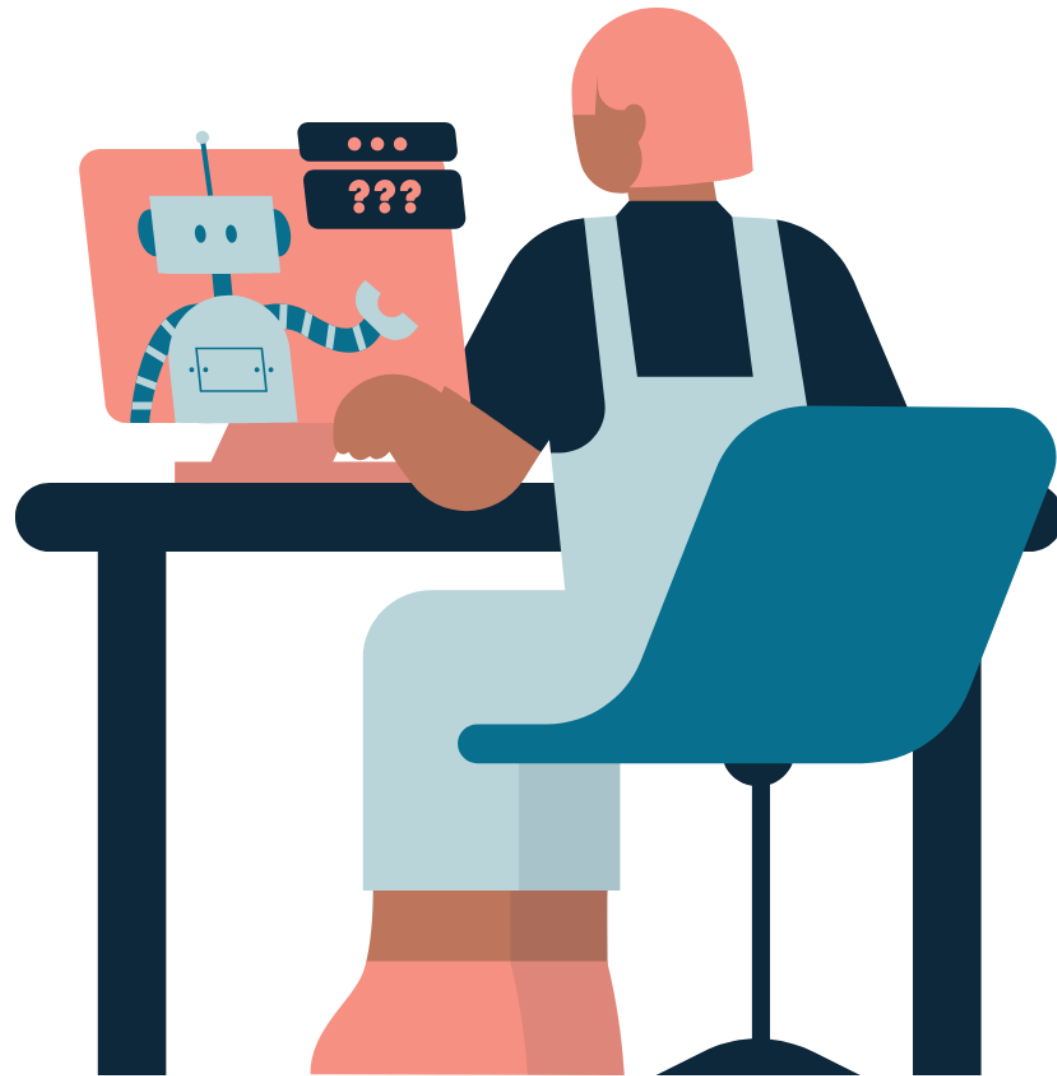
LlamaIndex

- Planning
- Use tools
- Remember information
- Work as a team



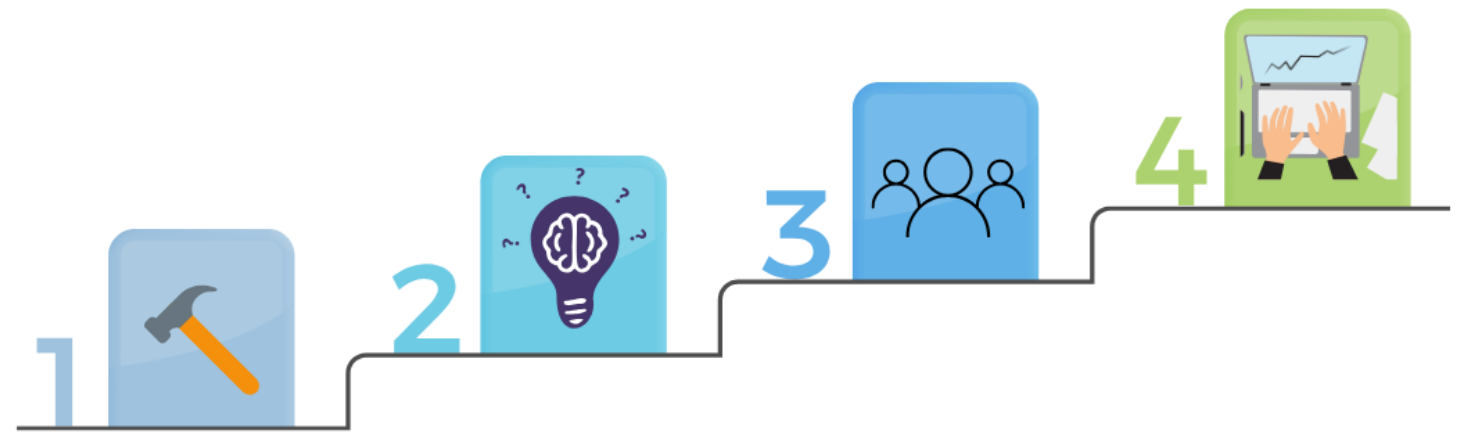
A research agent

- A deep research agent like the most advanced ChatGPT models



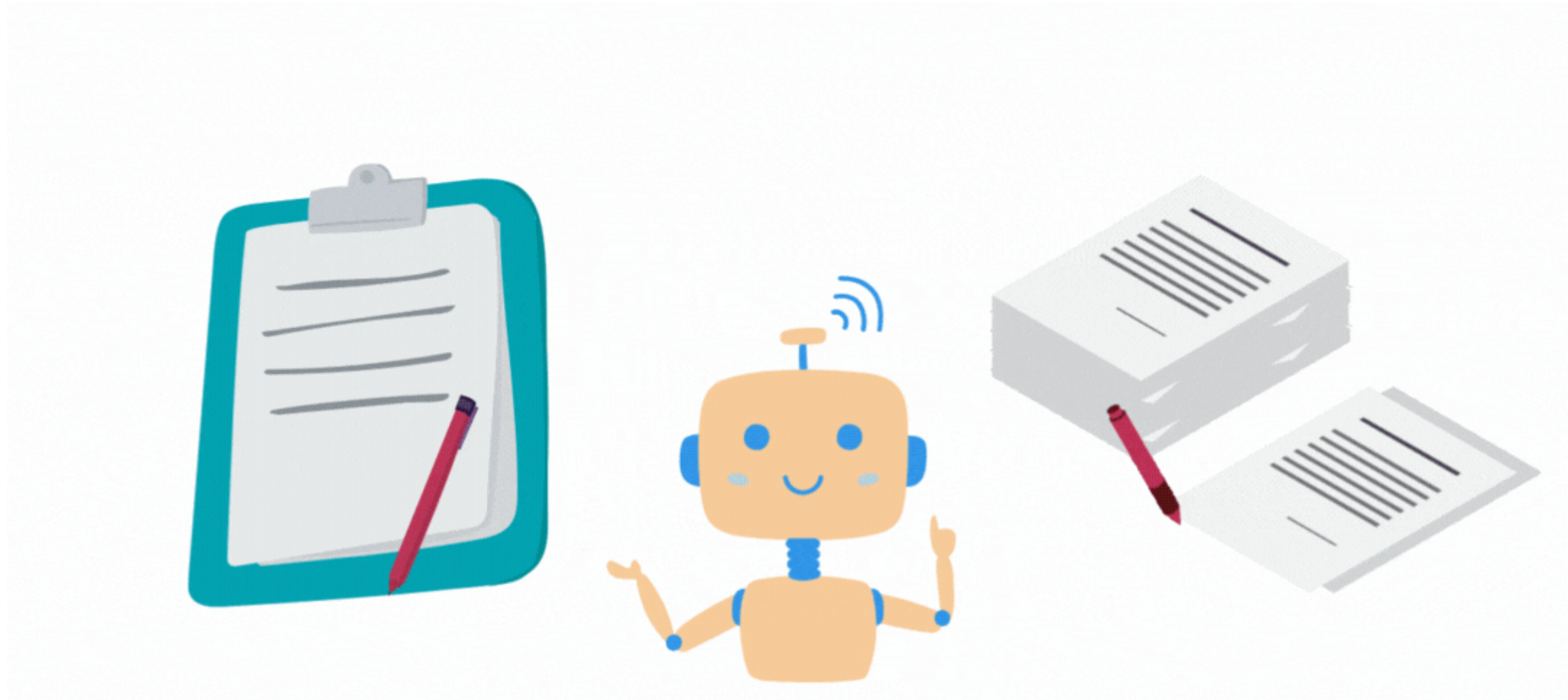
Step-by-step workflow

- Gradual build-up:
 - Tool use
 - Memory
 - Multi-agent workflows
 - Human checkpoints



Agents to power your work

- Market reports
- Long-form content



Course format

Multi-agent System with AgentWorkflow

Why stop at one specialist when we can assemble an Avengers-level AI squad? This section shows how to spin up multiple expert agents that collaborate toward a single goal.

Our system will have three agents:

- A `ResearchAgent` that searches the web for information on the given topic.
- A `WriteAgent` that will write the report using the information found by the `ResearchAgent`.
- A `ReviewAgent` that reviews the report and provides feedback.

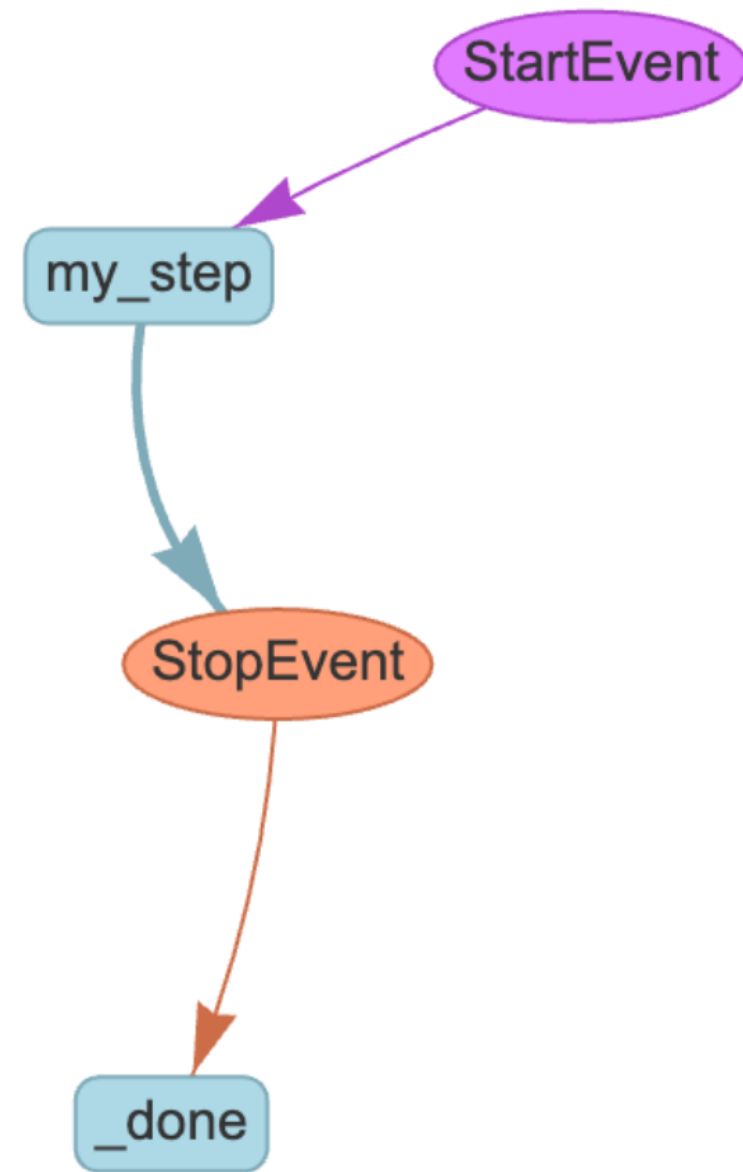
We will use the `AgentWorkflow` class to create a multi-agent system that will execute these agents in order.

There are a lot of ways we could go about building a system to perform this task. In this example, we will use a few tools to help with the research and writing processes.

1. A `web_search` tool to search the web for information on the given topic.
2. A `record_notes` tool which will save research found on the web to the state so that the other tools can use it.
3. A `write_report` tool to write the report using the information found by the `ResearchAgent`.

Course goals

- LlamaIndex's AgentWorkflow
- Orchestrating multiple agents using LlamaIndex



Prerequisites

- Can write Python functions
- Can work with APIs
- Interest in building agentic workflows

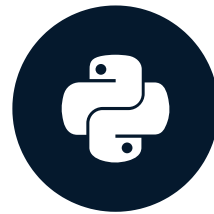


Let's practice!

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Getting started with LlamaIndex

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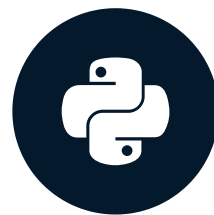
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What are agents?

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LLMs have limitations

- LLMs are brilliant but have limitations
- Answer with text responses from training
- Can't fetch info or act unless we provide new data



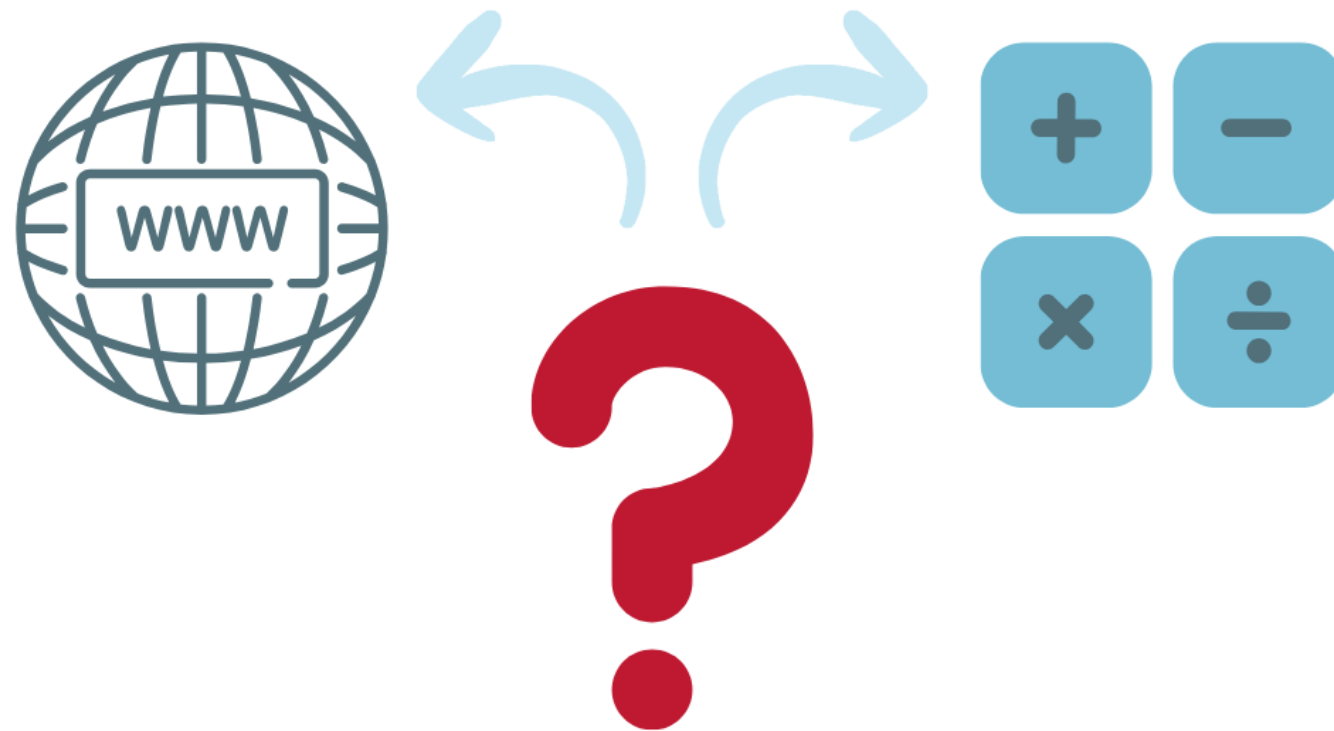
Agents: the bigger picture

- Leverage an AI model to interact
- It can:
 - Reason
 - Plan
 - Execute actions via tools



Agents: the bigger picture

- Scenario:
 - AI that can choose web search or step-by-step calculations
 - Handles complex or open-ended tasks
 - Breaks tasks into steps and uses tools



LLMs vs. Agents

- LLM:
 - If the task is simple

"Summarize this text..."
- Agent:
 - When task requires multi-step, external info, or conditional decisioning
 - When we want to add a tool

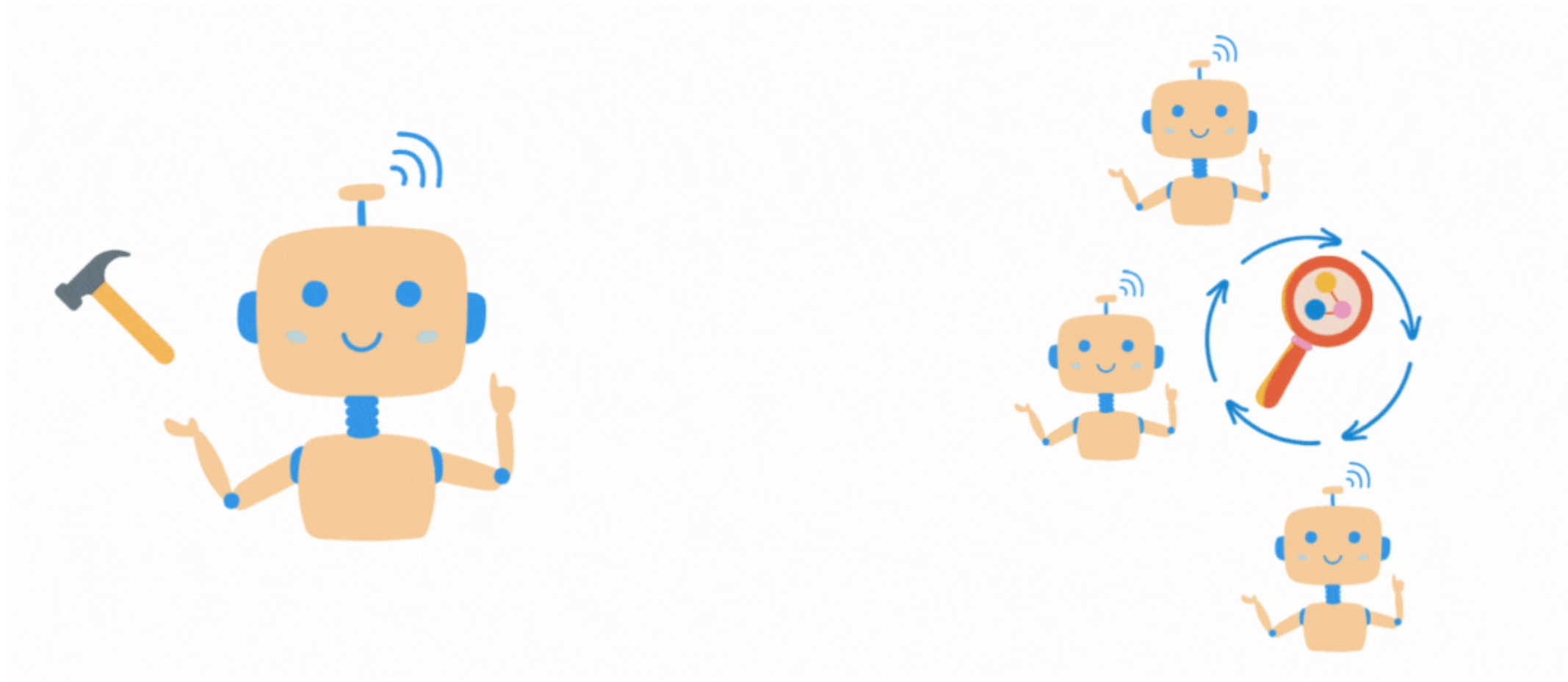
"Research this topic..."



Agentic workflows

Focus of the course:

- Agentic workflows
- From single agent + tool to deep research multi-agent systems

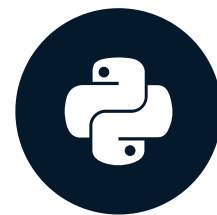


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Creating an agent

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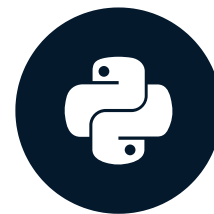
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Do agents remember?

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Stateless vs. Memory

Agents:

- Stateless by default
- Don't retain information between sessions
- Run independently each time



Memory matters

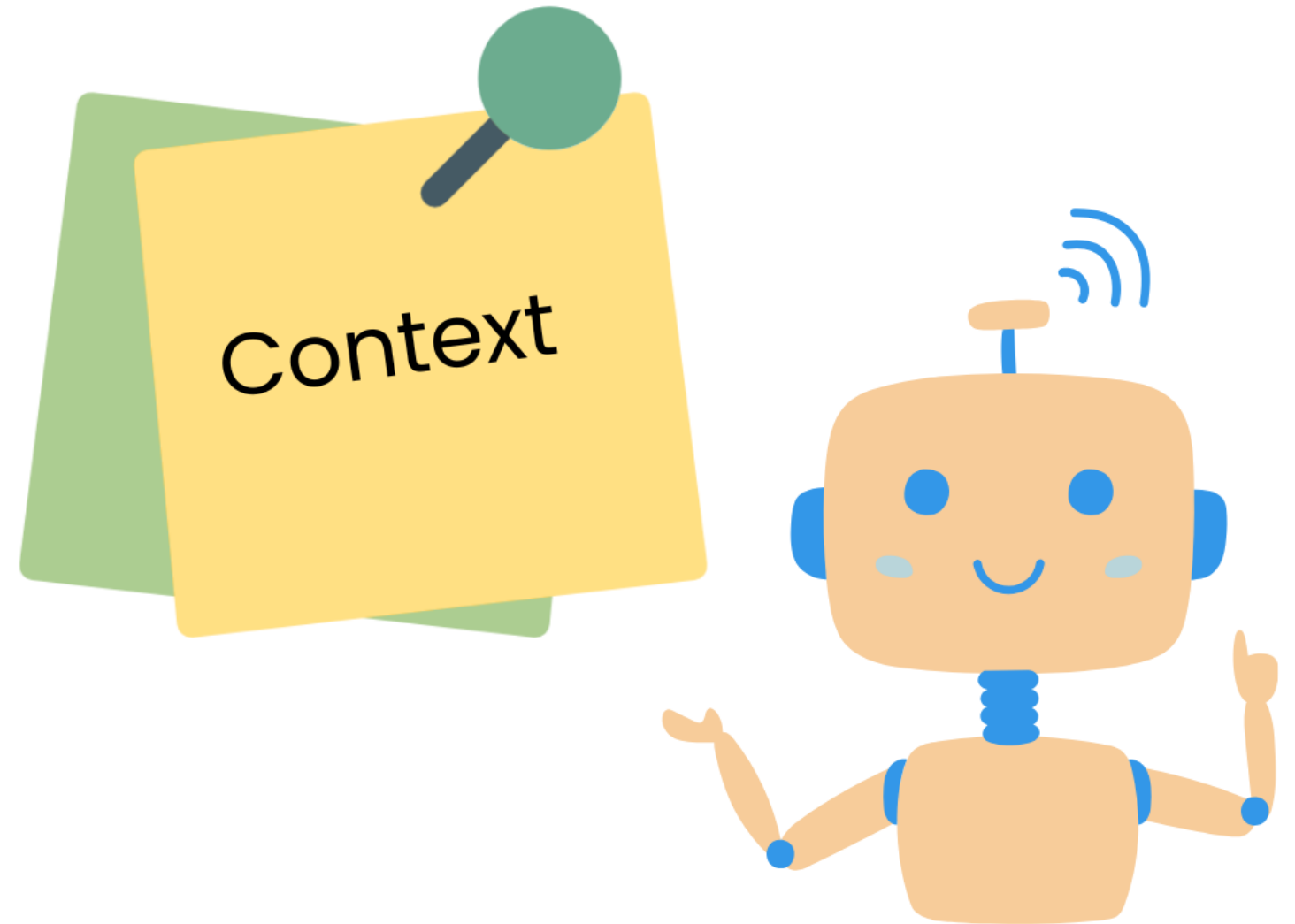
- Human memory carries conversations forward
- Agents need context to be helpful



Introducing context

- Example: remembering names
- LlamaIndex uses a Context object
 - Stores information between runs

Context: the agent's notepad or memory bank where it can store information to recall later.



From stateless to context-enabled

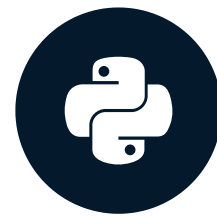
- Stateless agent:
 - Forgets everything after response
- Context-enabled agent:
 - Retains memory across chats
 - Needs to be enabled
- **Carrying information forward is key in workflows**

Let's practice!

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Maintaining state across agent runs

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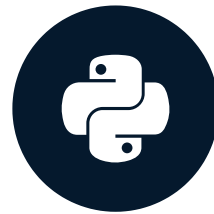
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Streaming output and events

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